

LEVEL 4 • CHALLENGE

Planet Hunt

Find the world made of starlight and carbon.

Tucked into Francis Benoni's notebook is a torn observation slip. No name on it — just a scatter of numbers and one underlined phrase: "the gem orbiting the crab. find what the astronomers chose to call it."

The slip describes a single, real exoplanet. Use its properties to pin down exactly which planet it is, then recover the official proper name that the International Astronomical Union assigned to it.

What the slip records about the planet:

- It is a super-Earth — a rocky world roughly twice Earth's diameter and about eight times its mass.
- Its year is astonishingly short: it completes one orbit in only about 18 hours.
- It is carbon-rich, which earned it a famous gemstone nickname in the press.
- Its host star is a naked-eye, Sun-like star about 40 light-years away, in the constellation Cancer (the Crab).
- It was first detected in 2004, and is the innermost of its star's several planets.

Database	NASA Exoplanet Archive — exoplanetarchive.ipac.caltech.edu
Also useful	a plain web / image search for the nickname
Flag format	CTC{propername} — one lowercase word, the planet's IAU proper name

Note: the flag is not the catalogue designation and not the nickname — it is the formal proper name the IAU approved for this planet.